

Microgrids are gaining popularity as a reliable solution for keeping lights on in communities, commercial operations, and utilities, with energy storage being a natural fit when paired with renewables. Additionally, energy shifting, where excess solar energy produced during the day is stored for use at night, is emerging as a significant application for energy storage.

Equipment and materials suppliers can play a pivotal role in developing lower-cost energy storage technologies. By forming development partnerships or supply agreements with startups or technology developers, these suppliers can offer alternative materials or advice to reduce costs. Their involvement can provide pathways towards more affordable energy storage solutions, addressing one of the critical challenges facing early-stage technologies.

EV batteries in fixed energy storage

The integration of electric vehicle (EV) batteries into fixed energy storage systems, commonly known as vehicle-to-grid (V2G) technology, presents a range of opportunities and challenges. This approach allows EV batteries to power homes or export energy to the grid, providing a valuable asset for grid resilience. It is applicable not only to residential customers but also to commercial fleets, offering substantial energy storage capacity.

However, this integration requires buy-in from automakers to incorporate bidirectional charging capabilities in their vehicles, allowing energy to be both stored and discharged. Additionally, software and visibility into users' energy usage are necessary to optimise the utilisation of the EV battery.

Recycling plays a crucial role in the life cycle of EV batteries, impacting sustainability. While the process can vary, car manufacturers see

value in recycling to extract valuable materials from batteries for resale. Although driven more by material recovery than circularity, recycling contributes to sustainability by reclaiming plastics, circuitry, and metals like aluminium and copper.

Used EV batteries are assessed for second-life applications through visual inspection and cycling tests to determine their state of health. Batteries with capacity below a certain threshold are disqualified. Those that pass undergo further testing to determine their suitability for second-life energy storage, where their performance is carefully managed to ensure safe operation.

Capacity degradation in EV batteries affects their performance in fixed energy storage applications. As capacity decreases, voltage characteristics change, requiring intelligent management of the remaining capacity. Second-life battery systems often throttle performance to operate safely within these constraints.

The repurposing of EV batteries for energy storage systems is governed by safety standards such as UL 1974 certification. Thermal management is crucial in both EV and stationary storage systems to mitigate external environmental effects. EVs use passive and active cooling, while stationary storage relies on HVAC systems to maintain optimal temperatures.

The growing stockpile of EV batteries with residual capacity is driving their use in fixed energy storage systems. This market is expanding as companies seek to monetise second-life batteries in short-duration energy storage applications.

The effective use of EV batteries in stationary energy storage is not solely technology-driven but is influenced by the increasing volume of end-of-life EV batteries. Early attempts at repurposing, such as Toyota's work with Prius batteries,

were deemed unviable due to low volumes. However, the expanding EV market has made second-life applications more feasible, with software and testing protocols evolving to ensure safe and efficient operation.

The integration of renewable energy sources like wind and solar power into the power grid necessitates innovative solutions to address their inherent intermittency. Energy storage systems, smart grid technologies, and demand response plans are key strategies that help maintain grid reliability and balance supply and demand. These approaches enable the efficient management of energy resources, ensuring that renewable energy can be effectively integrated into the grid without compromising stability. As the share of renewables continues to grow, the development and deployment of these technologies become increasingly vital for a sustainable and reliable energy future.

Emerging trends and technologies in energy storage, such as sodium-ion batteries, offer promising avenues for further enhancing the integration of renewables into the grid. While sodium-ion batteries may have a lower efficiency compared to lithium-ion, their potential for lower cost materials and similar manufacturing processes make them an exciting area of development.

Ultimately, reducing costs remains a critical goal across all energy storage technologies, as affordable solutions will be key to widespread adoption and a successful transition to a renewable energy-dominated grid. **EFY**

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